

Ruijia (Hannah) Guan

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Education

University of Michigan

BS, Cognitive Science & Computer Science

GPA: 3.94/4.00

Ann Arbor, MI

2022 – 2025

University Honors

Colby College

Psychology & Computer Science (Transferred to UMich)

GPA: 4.08/4.00

Waterville, ME

2020 – 2022

2021 Dean's List

Fields of Interest

Human-Computer Interaction; Social Computing; Cognitive Psychology; Collective Intelligence; Human-Centered AI; Mixed Reality;

Peer-Reviewed Publication

- **Guan, R.**, Lee-Robbins, E., Wang, X., & Adar, E. ([Accepted](#)) VisQuestions: Constructing Evaluations for Communicative Visualizations. *2025 ACM Conference on Human Factors in Computing Systems, Extended Abstracts (CHI EA)*
- Esterwood, C., **Guan, R.**, Ye, X., & Robert, L. P. ([Accepted](#)). Virtually the same or realistically different: Do real-world robots differ from their virtual representations? *2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI)*.
- Johnson, J., Peralta, M., Kaur, M., Huang, R. S., Sheng, Z., **Guan, R.**, Rajaram, S., & Nebeling, M. ([Accepted](#)). Exploring MR-enhanced collaborative AI: Eliciting team perceptions and design considerations for the future of work. *2025 ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW), October cycle*.
- Ashkinaze, J., **Guan, R.**, Adar, E., Budak, C., & Gilbert, E. ([Accepted with Major Revision](#)). Seeing like an AI: How LLMs apply (and misapply) Wikipedia neutrality norms. *2025 ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW), July cycle*.
- Huffman, D., & **Guan, R.** ([Accepted, In Press](#)). Computational models suggest that human memory judgments exhibit interference due to the use of overlapping representations. *Psychological Review*. <https://doi.org/10.1037/rev0000517>

Working Projects

- **Cognitive Science Honors Thesis:** Structuring Strategic and Proactive AI Agents for Collective Deliberation.
- Yu, Z., Liu, Y., **Guan, R.**, & Vermillion, C. Large Language Model-driven Mixed Strategy Framework for Social Navigation. *Submitted to 2025 International Conference on Intelligent Robots and Systems (IROS)*

Research Experiences

[Social Computing]

Seeing Like an AI: How LLMs Apply (and Misapply) Wikipedia Neutrality Norms

Advisors: [Prof. Eytan Adar](#) and [Prof. Eric Gilbert](#)

Sep. 2023 – July. 2024

- Coded to generate LLM revision on biased Wikipedia texts and explored the outcomes with different prompting techniques including zero-shot prompting and constitutional AI chain; analyzed the LLM revision results with metrics including edit distances and precision-recall scores to measure the quality and investigate the differences with the human edition
- Co-designed the user study to evaluate the human vs. LLM modified versions, collected Prolific data (N=147), and co-wrote the paper as the second author. [Preprint: https://arxiv.org/abs/2407.04183](https://arxiv.org/abs/2407.04183)

[Collective Intelligence]

DISCO: Designing Intelligent Spaces For Collaboration

Advisors: [Dr. Janet Johnson](#) and [Prof. Michael Nebeling](#)

May. 2024 – Present

- Designed and developed six AI agents around the theme of "Smart Cities," utilizing vector database-based RAG technology for dynamic knowledge support; Built an interactive full-stack application using React and Flask to integrate intelligent agent management, server-Unity data interaction, text-to-speech conversion, and real-time conversation analysis
- Contributed to the design of MR collaborative spaces, supporting experiments with diverse agent appearances in Unity and exploring the design of various collaborative space elements
- Conducted workshop studies on 6 professional teams, submitted the exploratory paper to CSCW 2025, and working on the revised Agent pipeline as the Cog Sci honors thesis

Virtually The Same or Realistically Different: Do Real-World Robots Differ From Their Virtual Representations

Advisor: [Prof. Lionel Robert](#)

May. 2024 – Present

- Developed the systematic review research plan including literature review strategies and paper screening criterias

- Conducted the paper coding via *Ryaan* and *Notion*, undertook the part of the follow-up meta-analysis via *R* to analyze the effects of robot's physical embodiment on anthropomorphism, social presence, and user engagement
- First-stage paper accepted by HRI 2025; Preparing a journal version with more outcome metrics and alignment with the HRI analysis taxonomy, specifying field of application, task specification, and human-robot team composition

VisQuestions: Constructing Evaluations for Communicative Visualizations

Advisor: [Prof. Eytan Adar](#)

Feb. 2023 – Sep. 2024

- Developed a web application that generates multiple-choice questions based on selected text and the learning objective framework, facilitates the visualization project management, and streamlines the evaluation test deployment and analysis
- Co-conducted the system usability study with 9 participants and collected Prolific data (N=1096) to analyze the quality of system-generated tests

[Cognitive Psychology]

Computational Models on Human Memory Judgments

Advisor: [Prof. Derek Huffman](#)

Jun. 2021 – May 2022

- Coded the online experiment (N=153) that runs mnemonic similarity tasks to investigate the pattern separation of hippocampal function; analyzed the experiment data and co-analyzed the comparison among behavioral results and results from common computational hippocampus models
- Wrote sections of behavioral experiments and one hippocampus model; paper submitted to *Psychological Review*

Selected Project Experiences

Fine-tuning and Local Deployment of a Large Language Model (LLM) based on the AMI Corpus

Feb. 2025 – Mar. 2025

- Processed the AMI meeting corpus to complete data cleaning, hierarchical extraction of high-quality dialogues, and construction of instruction-input-output training formats
- Fine-tuned LLaMA-3 8B on Google Colab using Unsloth + QLoRA, leveraging the curated dataset to enhance the model's strategic dialogue capabilities in meeting scenarios
- Converted the model to GGUF format and deployed it using llama.cpp + FastAPI, providing low-latency API access

Analyzing Emotional Expression and Engagement Dynamics in Reddit Communities

Oct. 2024 – Dec. 2024

- Extracted posts, comments, and replies in 8 subreddits with 33719 unique users with Python's PRAW API
- Applied NLP techniques to analyze online community dynamics, including word frequency analysis, sentiment analysis (TextBlob), emotion detection (DistilBERT), and SBERT-based comment classification; Conducted time-series analysis to track the impact of user interactions on sentiment and subsequent behaviors
- Report link: https://www.ruijia-hannah-guan.com/_files/ugd/a739a9_3ebc9367ab9640cdaadf116d751ded49.pdf

RepetitionVR: A VR Simulator to Overcome Social Anxiety and Isolation

April 2024

- Co-developed a VR narrative project as part of the U-M course '*Extended Reality and Society*', explored diverse social scenarios and constructed multiple social practice in Unreal Engine
- Contributed to design motivation analysis, background interviews, development logs, and trailer production
- Project showcased on the official website: <https://sites.google.com/umich.edu/repetvr/home>

Interactive Exhibition: Worldwide Tea

Sep. 2023 – Feb. 2024

- Project under *ArtsEngine, Multidisciplinary Design Program, UMich*; Served as Engineering Sub-team Leader
- Collaborated in designing an interactive tea culture exhibition, contributing to space layout and user scenario development
- Designed the prototype using Arduino triggers to interact with the front-end tea card presentation and video control; constructed and managed 28 instances in the 10-day exhibition. [Annual Expo Presentation Link](#)

Teaching and Working Experiences

University of Michigan

Ann Arbor, MI

Peer Facilitator, Weinberg Institute for Cognitive Science

Sep. 2023 – May. 2024

- Conducted 3 weekly office hours, providing students with insights into course registration and extra-curricular opportunities
- Co-plan community-building events and managed department resources including weekly email communications

Project Grader, Discrete Mathematics course

Jan. 2023 – May 2023

- Graded homework sets and facilitated course design

Colby College

Waterville, ME

Teaching Assistant, Honor Calculus I & II course

Sep. 2021 – May 2022

- Held office hours and graded student homework; facilitated improvement of course design and coursework

Incus Company Limited

Shenzhen, Guangdong, China

User Research Intern

Jun. 2021 – Aug. 2021

- Conducted statistical analysis and user behavior tracking on backend data from hearing aid products to identify usability bottlenecks and propose functional optimizations; Led multiple rounds of field studies with 60+ participants, employing mixed research methods (quantitative surveys and qualitative interviews) to extract actionable user insights
- Authored comprehensive research reports, recommending UI and interaction design improvements, and validated findings through sentiment analysis and A/B testing